

COMP1531

Software Engineering Fundamentals

Week 8

Design - Conceptual Modelling (Cont'd)

Recap

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- What is a Model?
- What is a conceptual model?
 - Structural Diagrams
 - Behavioural Diagrams
- State Diagrams
- State Machine



What is a Model?

— — —

A model is a **simplified representation** of a more complex system that helps us understand, communicate, or reason about that system.

Key Points

- Reduces complexity → focuses on important features
- Helps predict, analyse, or document system behaviour
- Used across many domains - from maths to software to aerodynamics

Why models matter

- Enable communication between stakeholders
- Provide a reference for design and implementation
- Support reasoning about current, past, or future system states
- Models are cheaper to create than building real systems.



Conceptual Modelling

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A conceptual model is a type of model that captures a system in a conceptual way rather than a physical way.

They tend to be **diagrammatic** or **visual**.

Which of the following are examples of Conceptual Models?

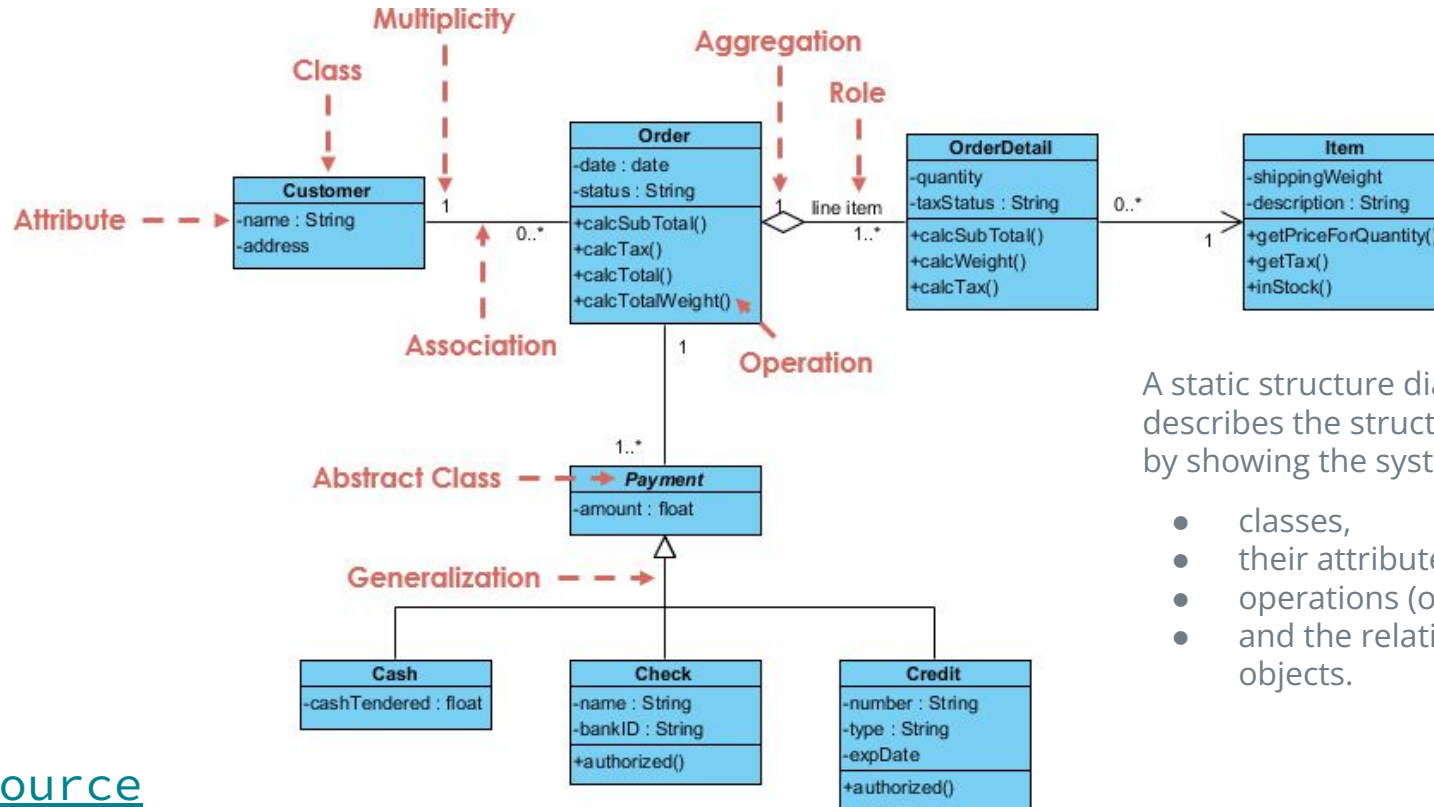
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1. **Mathematical models/functions**
2. **Data models:** To model databases
3. **Domain models:** To model components of a software system
4. **State transition models:** To model how a program operates at different stages
5. **Architectural scale models** - a miniature 3D printed model of a house
6. **Topographic (terrain) models** - e.g., a 3D model of mountains, rivers, or a city for urban planning
7. **Molecular models** - e.g., ball-and-stick models showing the structure of chemical compounds such as water (H_2O) or DNA.



Conceptual Modelling

Structural Models : 1. UML Class Diagrams (OO Programming)



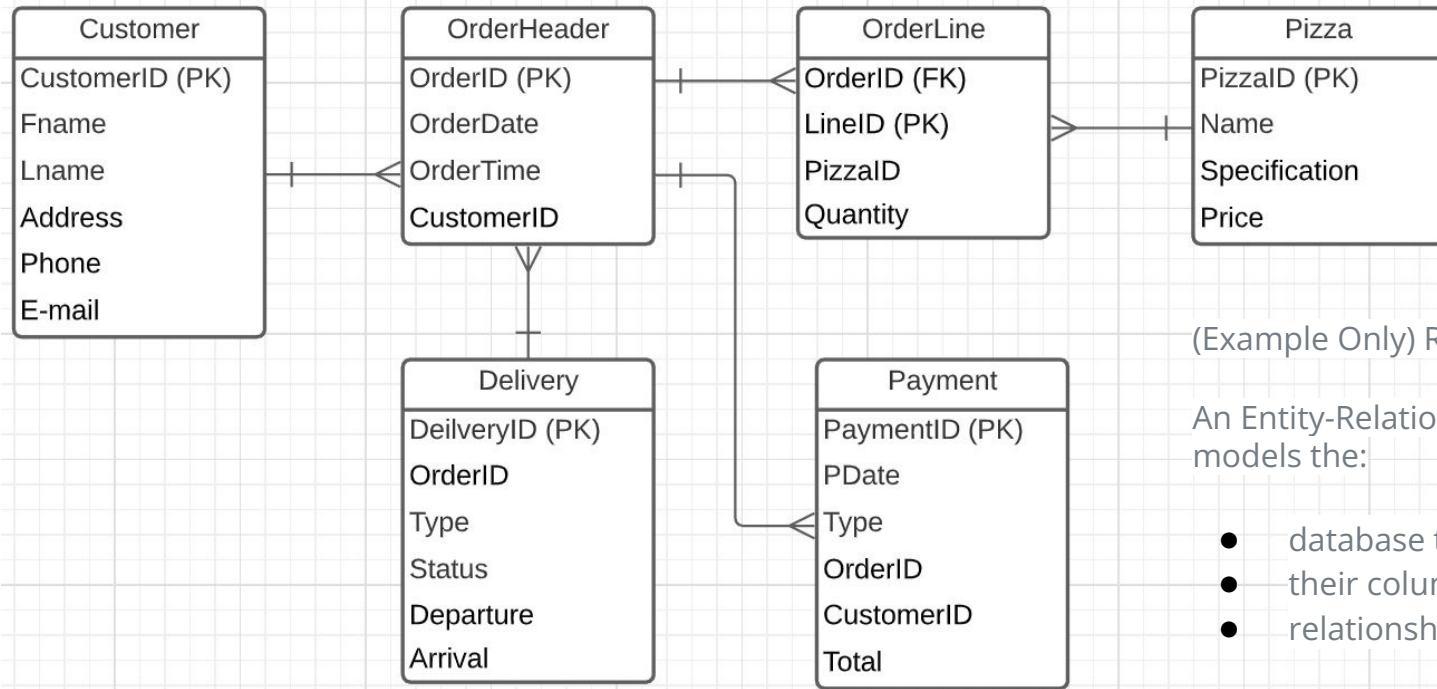
A static structure diagram that describes the structure of a system by showing the system's:

- classes,
- their attributes,
- operations (or methods),
- and the relationships among objects.



Conceptual Modelling

Structural Models : 2. ER Diagram (Database Design)



(Example Only) Restaurant Database

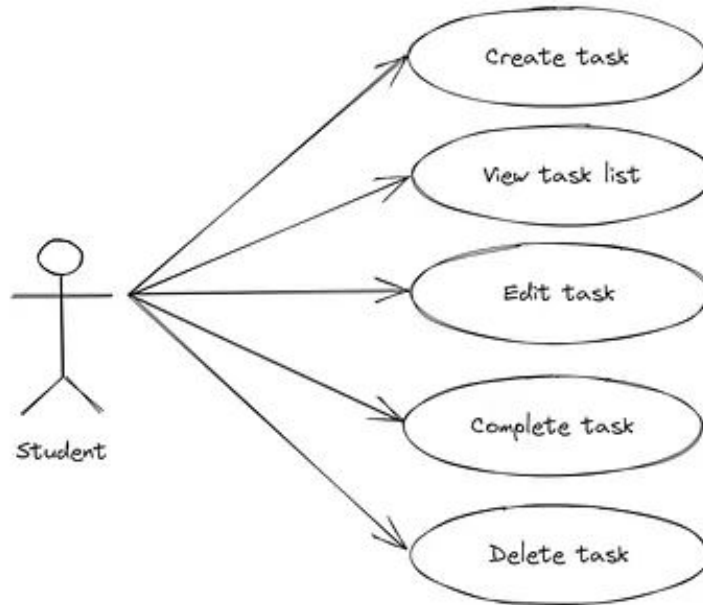
An Entity-Relationship (ER) Diagram models the:

- database tables,
- their columns, and
- relationships.



Conceptual Modelling

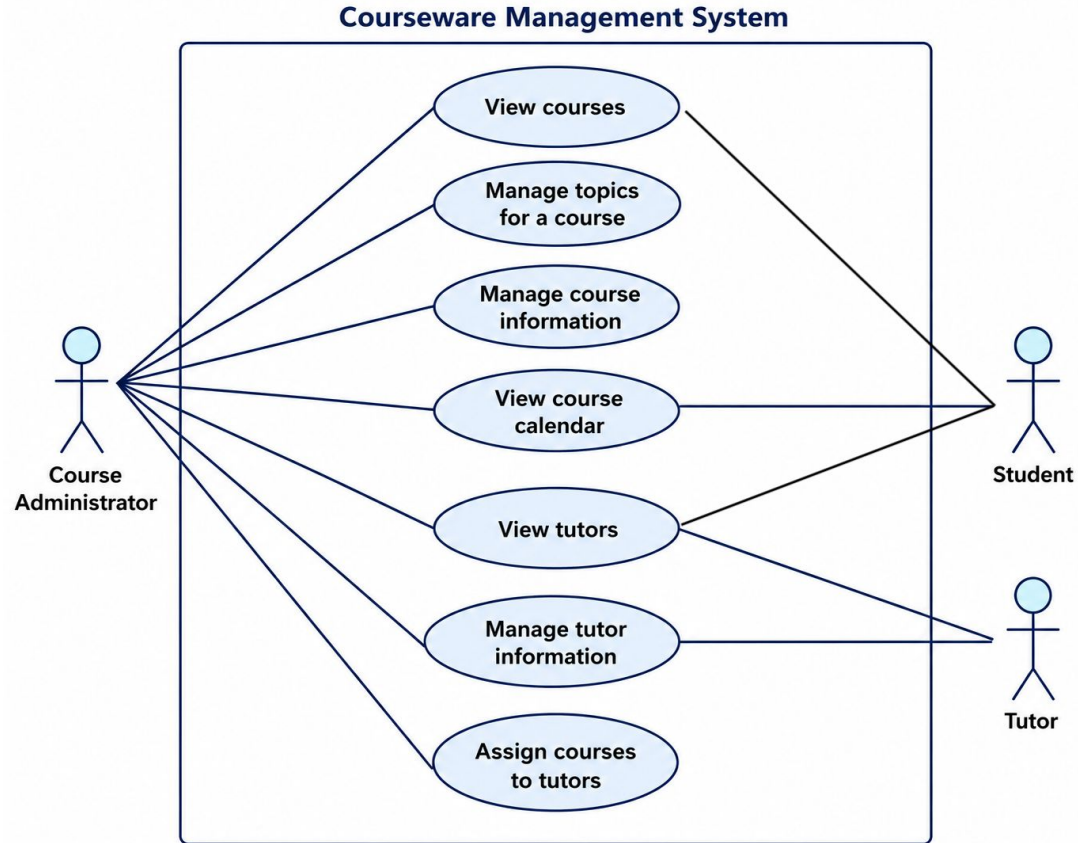
Behavioural Models : 1. Use Case Diagrams



The use case diagram presents **the actors** and the **actions** they can take when interacting with the system. CRUD operations are a good starting point to come up with use cases.

*Refer to: Lecture 2.5
(Slides 32 - 51)*

Behavioural Models : Use Case Diagram (Multiple Actors)



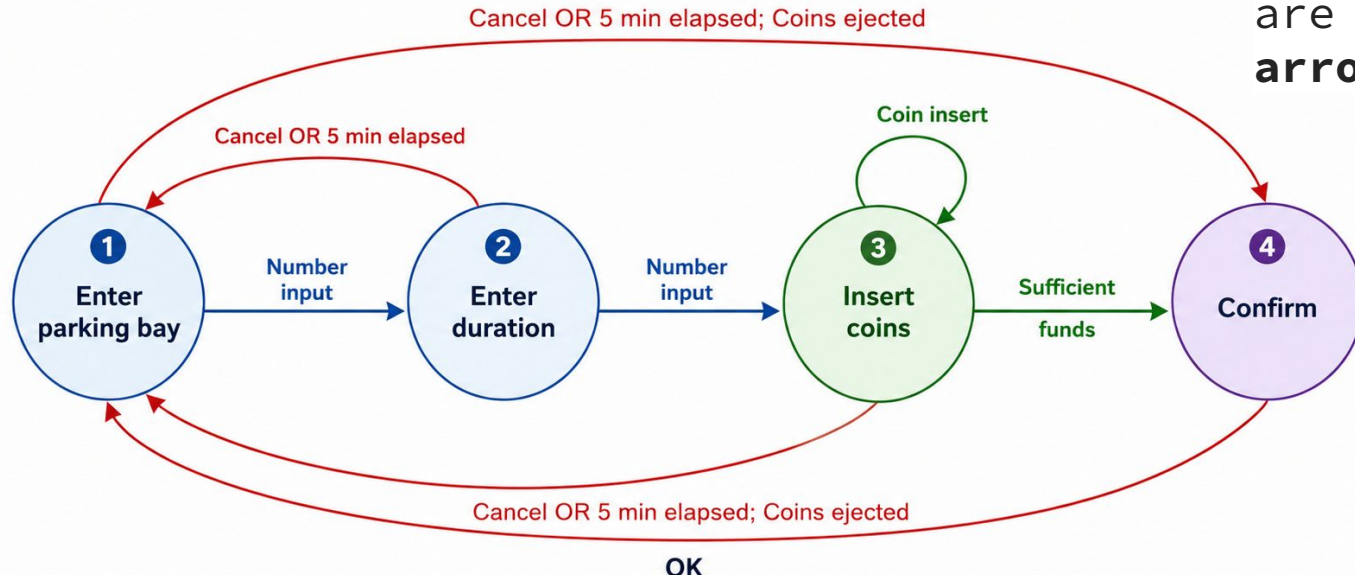


Conceptual Modelling

Behavioural Models : 2. State Diagrams



Parking Meter Example



A diagrammatic representation of a state. States are **circles**, transitions are labelled **arrows** connecting them

State Machine

A **state machine**, also called a **Finite State Machine (FSM)** is a **behavioural model** that describes **how a system changes from one state.**

A **State diagram** is a **visual notation** used to draw and communicate that state machine.

Examples include:

-  **Traffic lights**
-  **Parking meters**
-  **Elevators**
-  **ATMs**
-  **Automatic doors**
-  **Login/authentication systems**
-  **Mobile app screens**
-  **Game characters**
-  **Robots**
-  **Network protocols**

Quiz:

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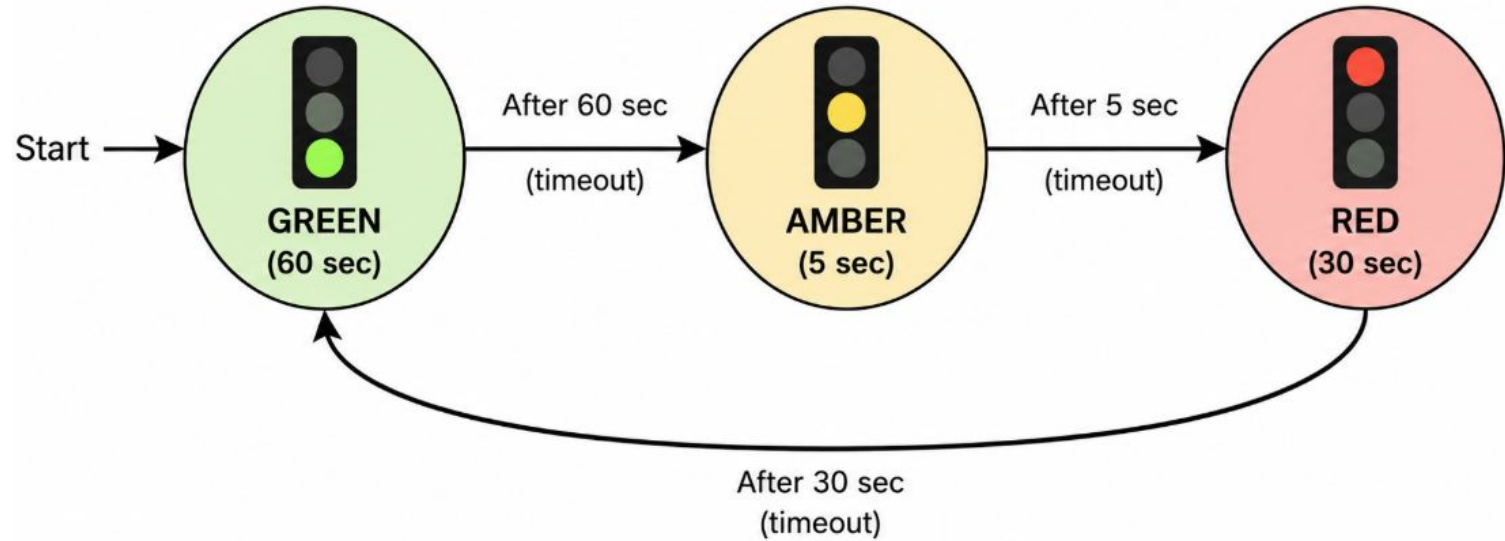


Question: Draw a **state diagram** for a traffic light controller. The controller cycles through the following states:

- **Red** for **30 seconds**
- **Amber** for **5 seconds**
- **Green** for **60 seconds**

After the Green state, the traffic light transits to Amber and then returns to the Red state and repeats the cycle indefinitely. Clearly label all states and timed transitions.

Solution:



*** What if car trips the sensor?*

Summary

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- Models simplify complex system ideas
- Conceptual models focus on understanding
 - **Structural models** show system components (architecture)
 - **Behavioural models** show system behaviour
- State machines define valid transitions (system states) in software
- Helps design before coding begins
 - Models prevent mistakes early → saves time, effort, and cost later.

Feedback

COMP1531 Feedback Survey

